

The Software Development Life Cycle (SDLC) in the age of AI Agents

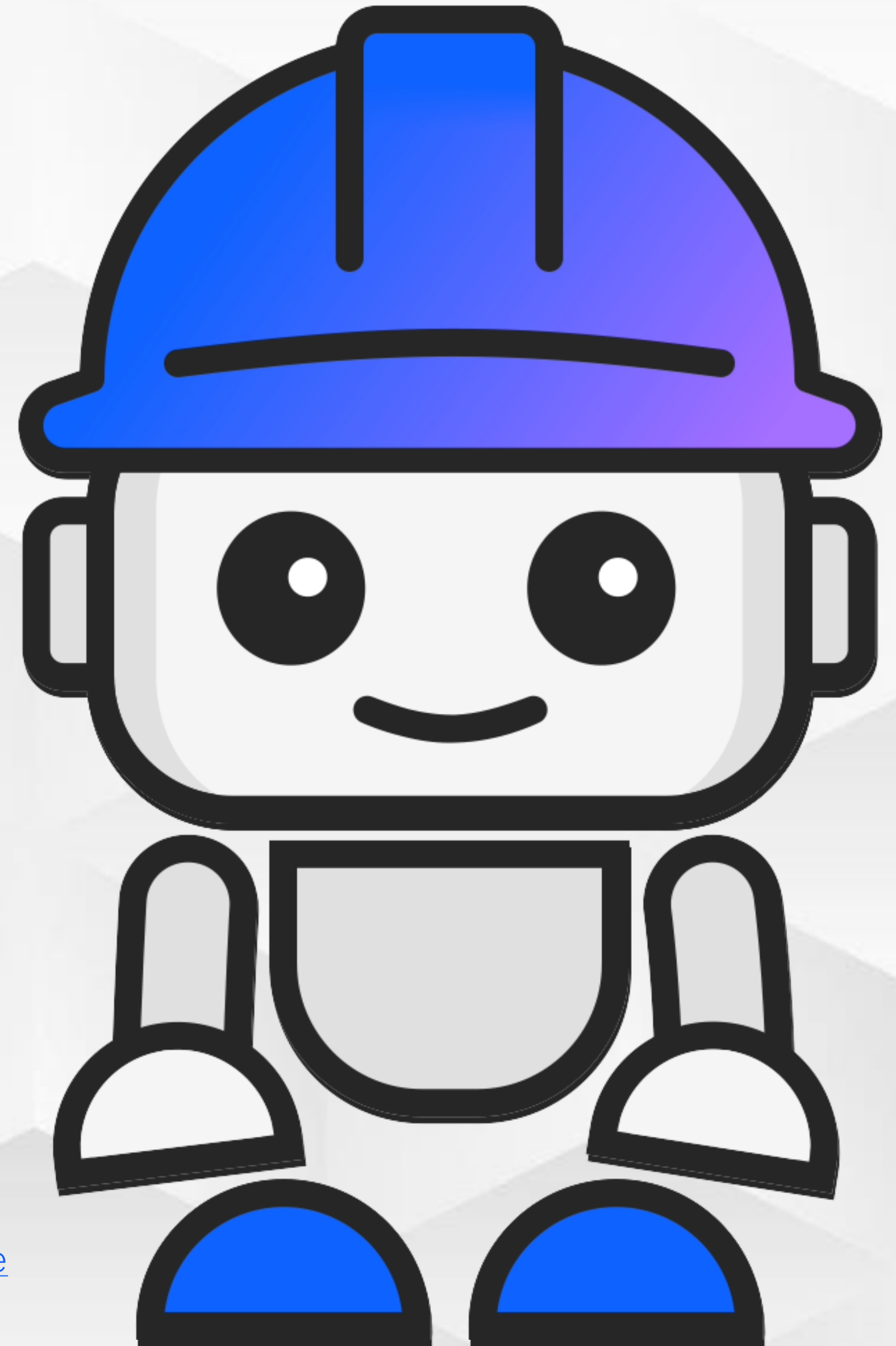
Harald Seipp (IBM)
Thomas Südbrocker (IBM)

QAware GmbH Cloud Native Night 09.06.2026

Repo Links:

<https://github.com/thomassuedbroecker/galaxium-travels-infrastructure-tsuedbro>

https://github.com/thomassuedbroecker/review_and_sdd_custom_ibm_bob_configuration_template



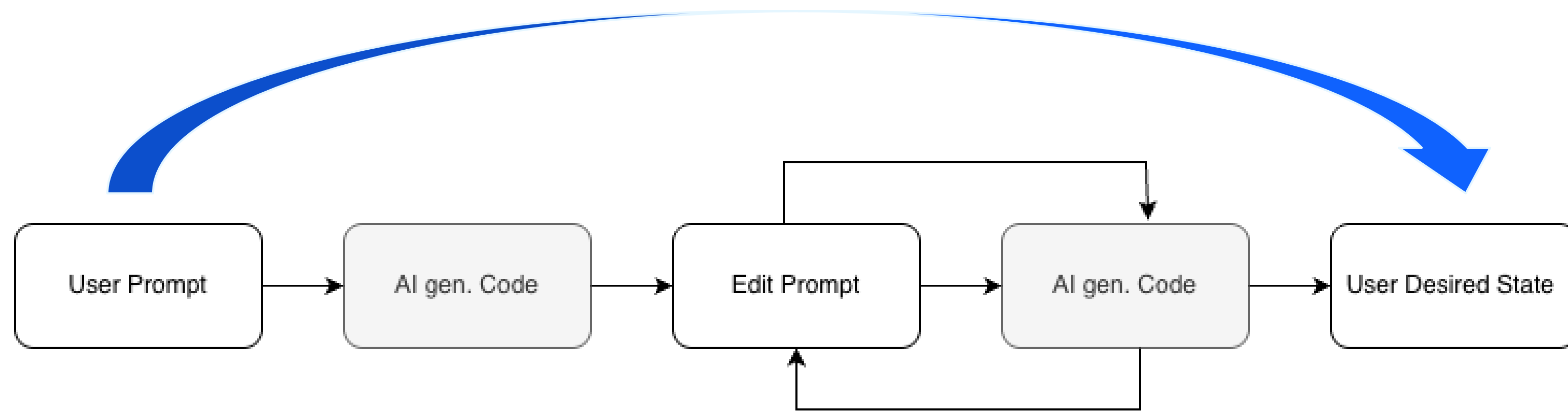
Opening Thoughts

AI already writes code better than most of us.

That is not the problem.
The problem is:

“Do we know when it is wrong?”

Vibe Coding



In February 2025, [Andrej Karpathy](#) invented the term vibe coding.

16:30 5G 54

u-de.codeengine.appdomain.cloud

AI Conference 2026
Impact Day - April 15, 2026

Official Agenda My Agenda

Official Agenda - Impact Day
Select sessions to build your personal agenda. Sessions are from 10:00 to 17:00 on April 15, 2026.

17 54 sessions available 0 selected

View My Agenda

Import Agenda

10:00 - 10:45 Masterclass

Funding Strategy for Research and Innovation projects in AI and Life Sciences

Daniel Niedermaier, Dr. Saskia Graf, Simon Glaser

Backstage

AI Agents & Automation

Add to My Agenda

10:00 - 10:45 Masterclass

The Development Shift

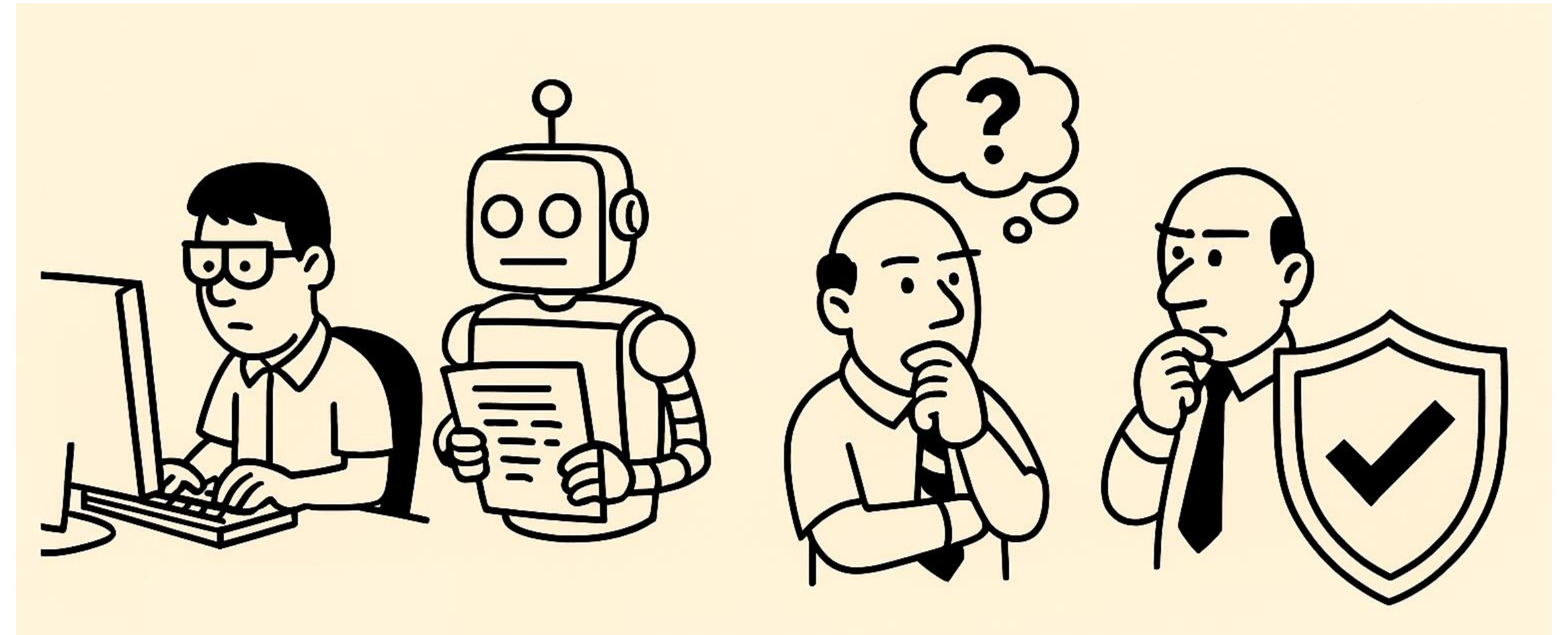
Old style

- Write code
- Review syntax
- Fix bugs late



New style

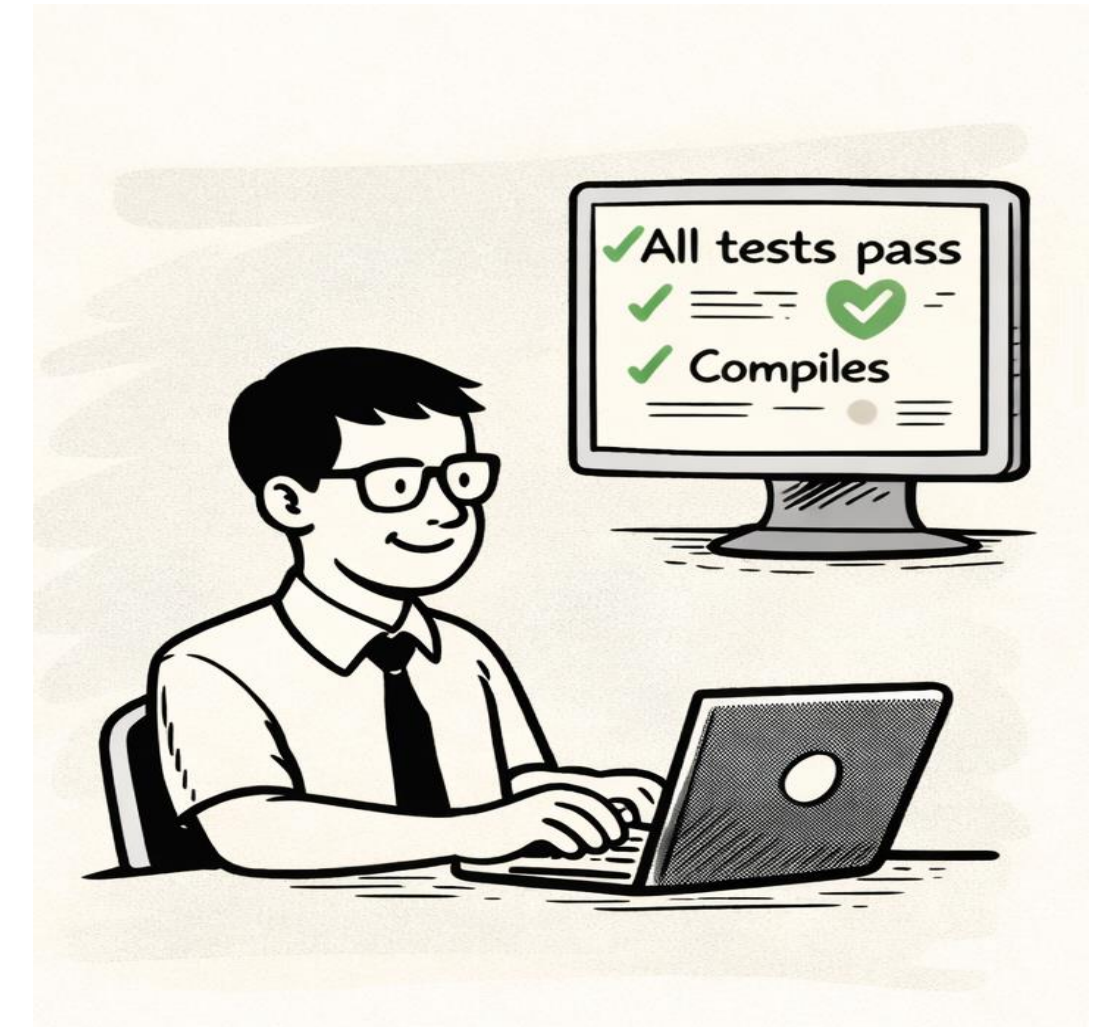
- AI implements "drafts"
- Humans review intent, logic, risk
- Security & compliance move left



Why Reviewing Is Hard

AI output:

- Looks correct
- Compiles



Real risks

- Often fails quietly → late and painful!
- Wrong assumptions
- Missing edge cases
- Sometimes uses fake data not even asked by the user

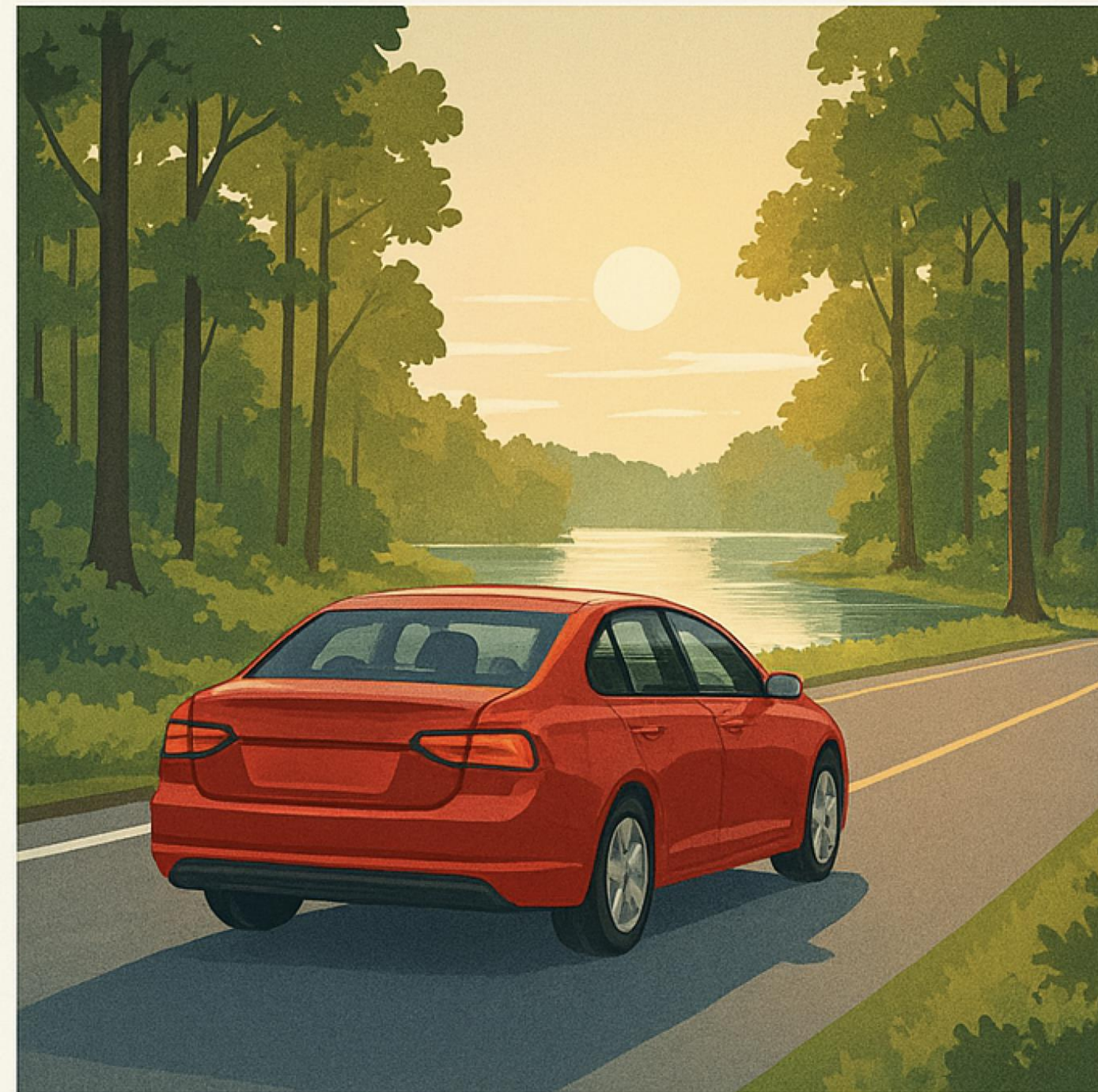


Did you say what you mean?

Is that just a first guess, or a clear, precise definition?

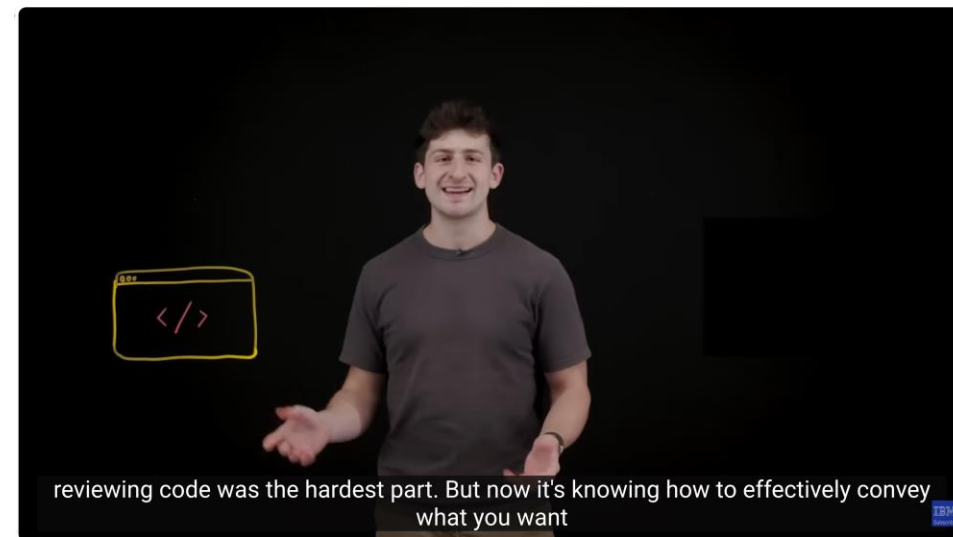
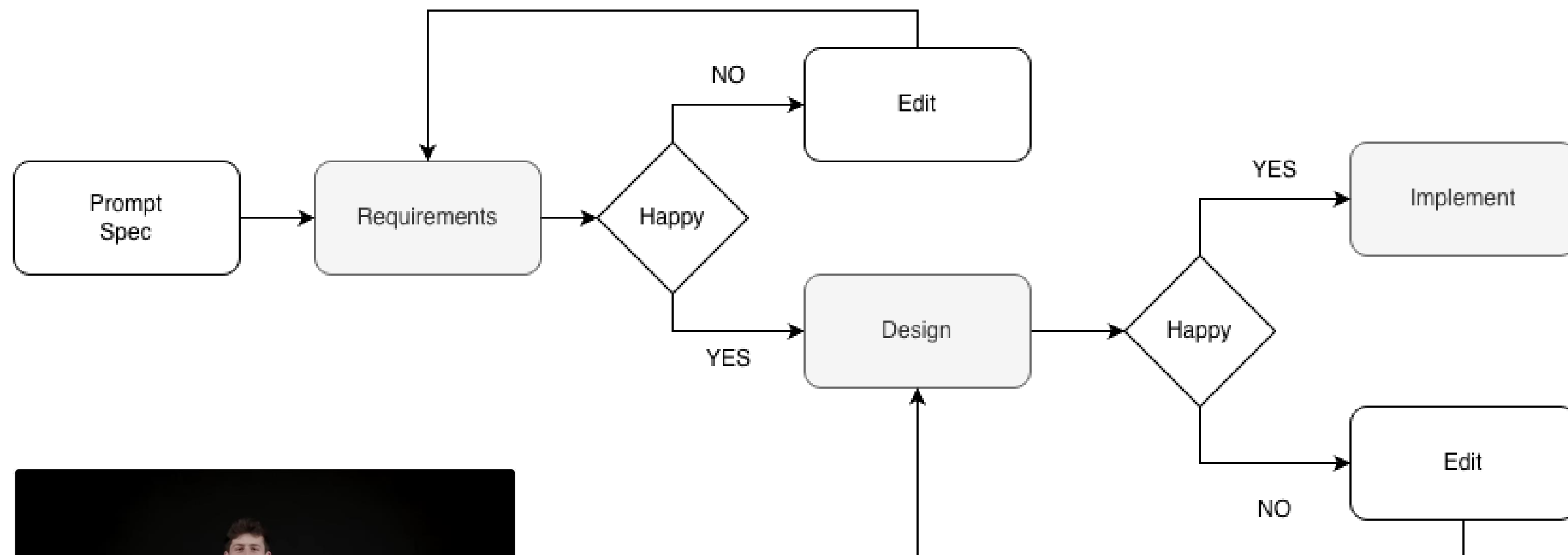


A car driving on a road.

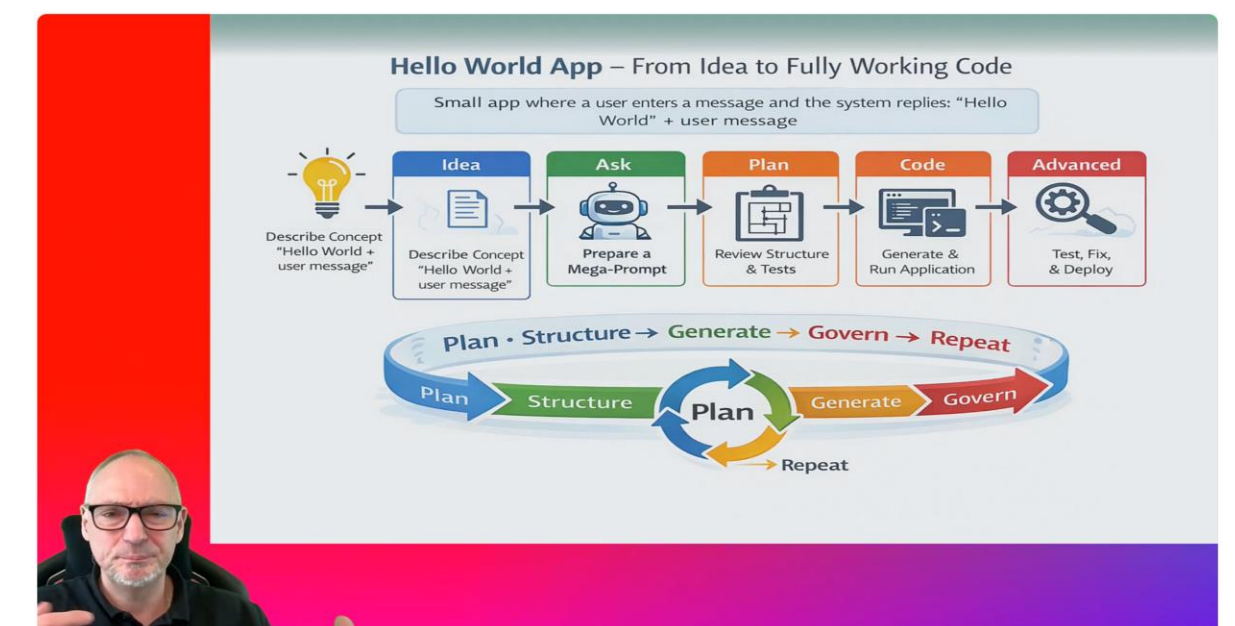


A red car driving on a road.
Along woods, by sunshine,
near a lake.

Spec-Driven Development



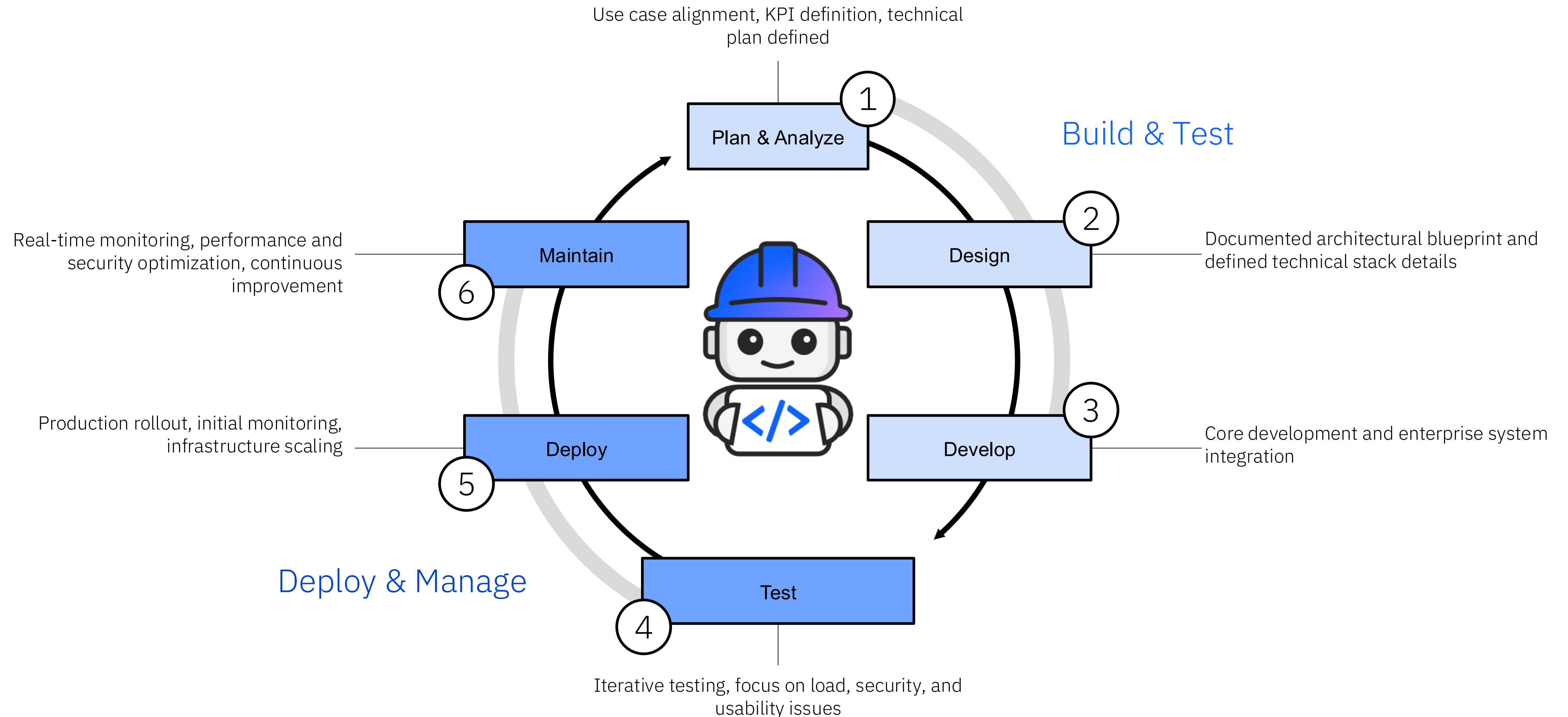
Resource - Spec-Driven Development: AI Assisted Coding Explained IBM Red Hat
<https://youtu.be/mViFYTwWvcM?is=Fy-iuy6yqlTJL4SY>



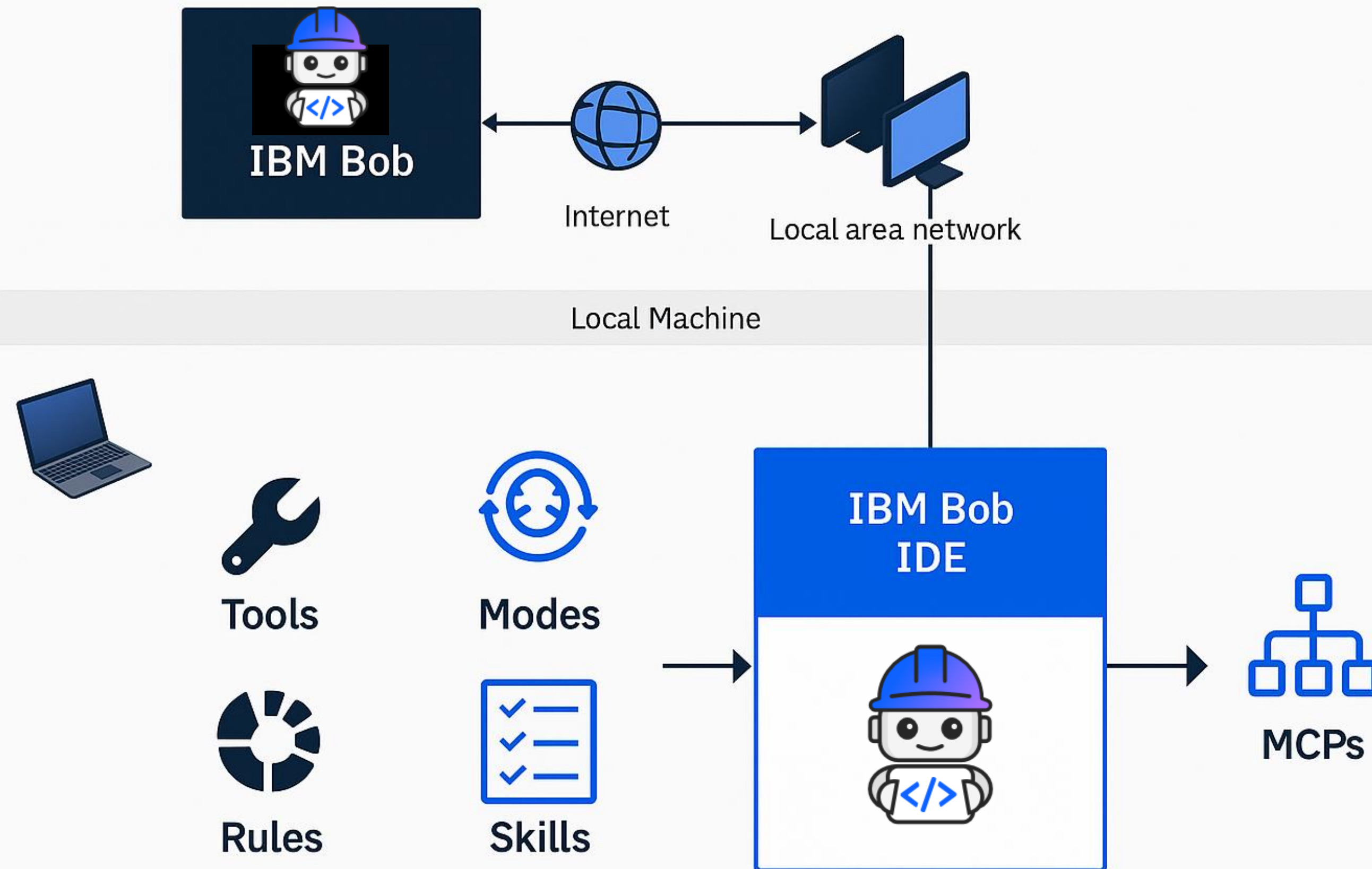
From Idea to Running App with IBM Bob – Prompt-Only Development (No Manual Coding) (25 min Version)
<https://youtu.be/c-AWDU-VO3Y>

Introducing IBM Bob, the AI partner across the SDLC

Delivering AI across the SDLC requires a different architecture. Bob is built for that.



IBM Bob components



Think about it like this:

- Modes = the roles
- Rules = the guardrails
- Skills = the playbooks
- Tools = the execution
- AGENTS.md = the team policy

From Weeks to Hours

Plan and Analyze

Traditional Workflow

- PM writes PRD in Jira (2-5 days)
- ... and more

Result: 3-6 weeks from idea to approved spec

Workflow

- PM describes feature in natural language to Bob
- Bob reads repo context (MCP) + existing stories for patterns
- ... and more

Result: 2-4 hours from idea to review-ready spec

Real Prompt example

```
$ bob
> I need to add a "Multi-Currency Payment Support" feature to the booking system.
> Context: Python FastAPI + SQLite, Flask web UIs, dual REST/MCP architecture,
> Keycloak OAuth + Basic Auth variants, Docker Compose deployment.
> Create a step-by-step implementation plan. Cover:
>   1. Database schema changes and migration strategy
>   2. Backend API modifications (REST + MCP endpoints)
>   3. Frontend UI updates for both web apps
>   4. Authentication flow impacts (OAuth + Basic Auth)
>   5. Docker Compose and deployment configuration
>   6. Testing strategy (unit, integration, WebUI matrix)
> Do NOT generate code yet. Output as a numbered checklist.
```

```
# Bob reads: README.md, QUALITY-CHECK.md, QUICKSTART.md,
# architecture diagrams, booking_system_rest/, booking_system_mcp/,
# galaxium-booking-web-app/, local-container/docker_compose.yaml
# Bob outputs: structured plan referencing YOUR existing patterns
# Human reviews, edits, approves → THEN coding begins
```

AI helping to review AI?

— fox guarding the henhouse or simply saving time?

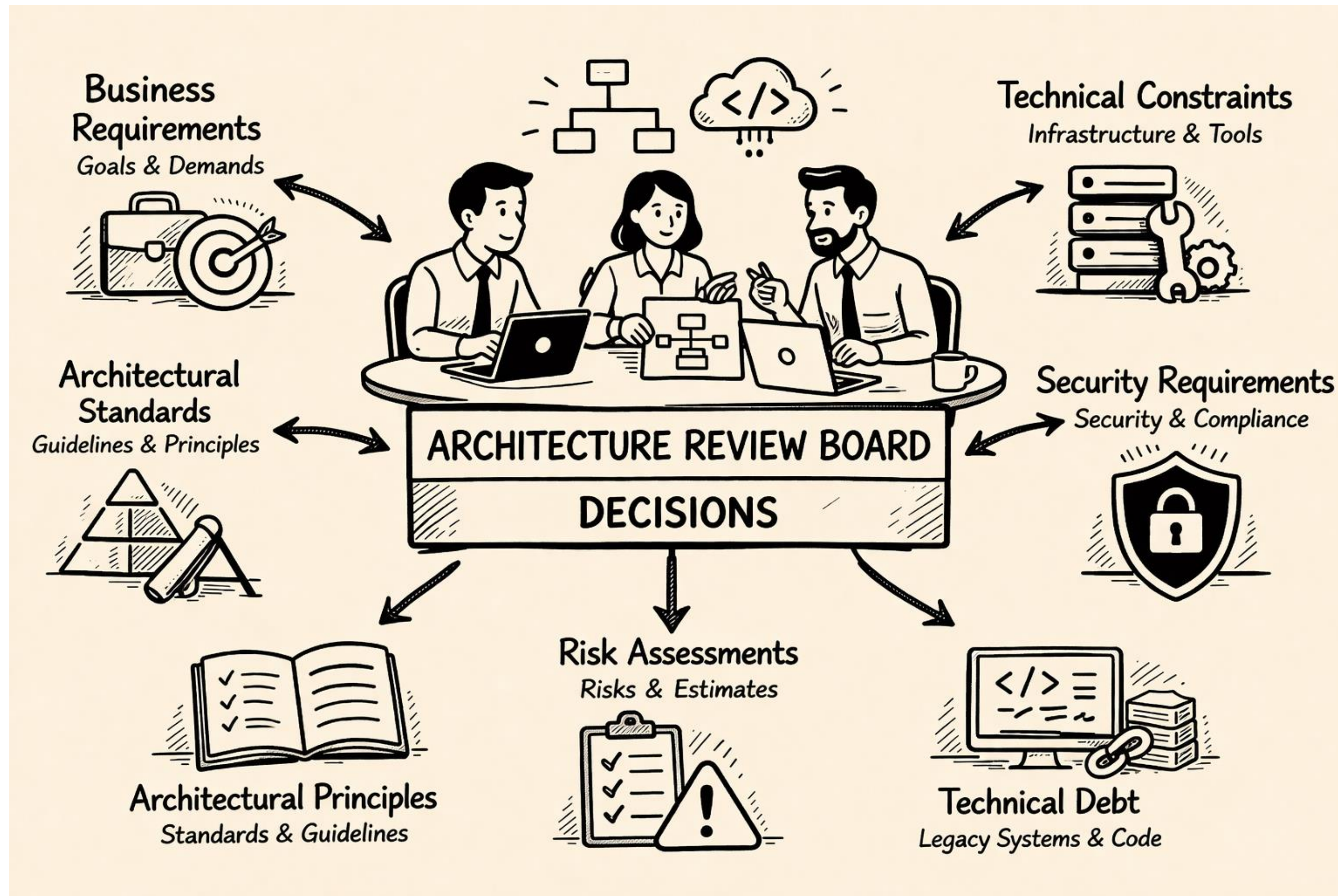


- We need to provide the right context and role in the modes and skills.

Think back to slide:

Did you say what you mean?

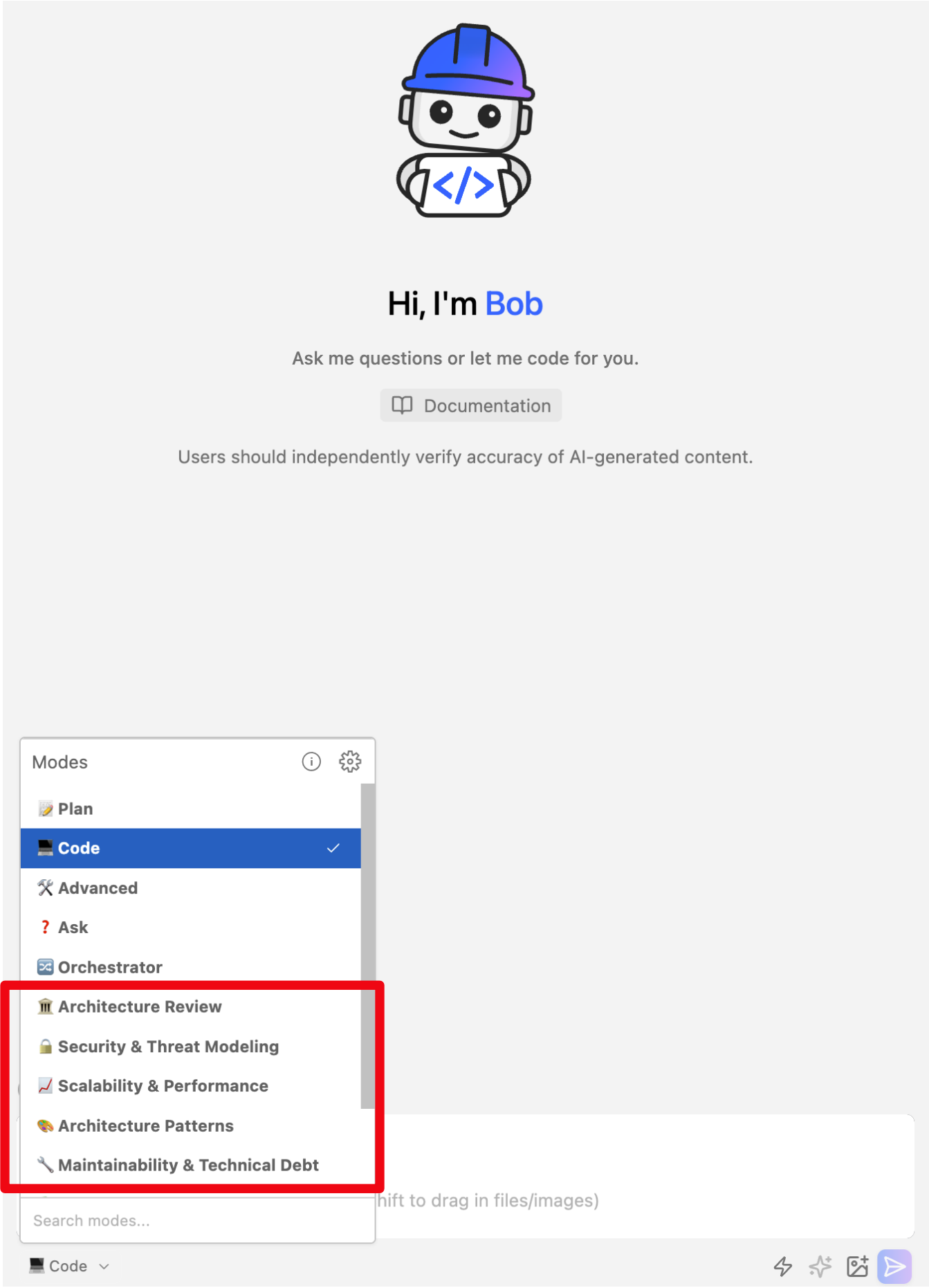
Architecture Review Board



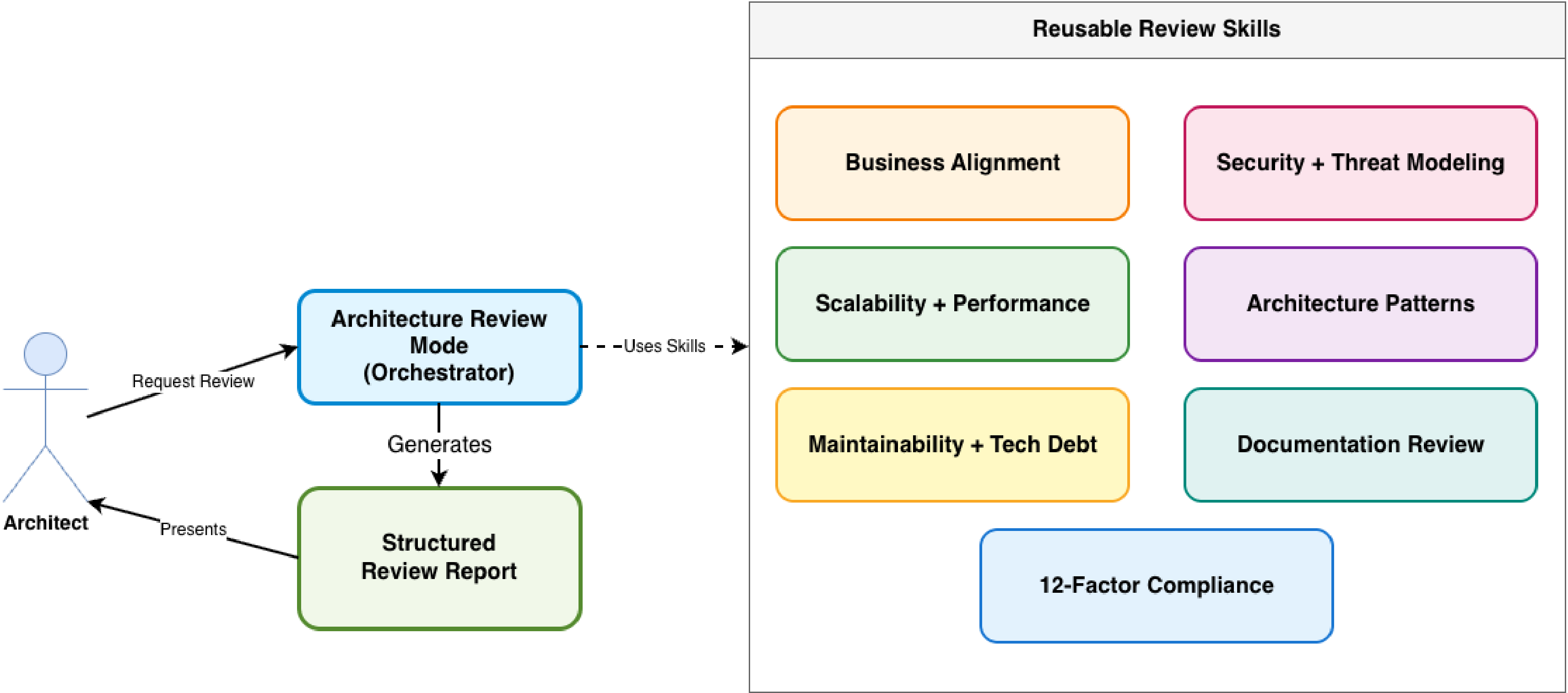
✓ galaxium-travels-infrastructure-tsuedbro

- > ai_generated_documentation
- > architecture
- > booking_system_mcp
- > booking_system_rest
- > deployment
- > galaxium-booking-web-app
- > galaxium-booking-web-app-mcp
- > HR_database
- > images
- > local-container
- > test-results
- > testing
- ◆ .gitignore
- 🔑 LICENSE
- ➡ manual_auth_check_using_the_commandline.md
- ➡ QUALITY-CHECK.md
- ➡ QUICKSTART.md
- 📘 README.md
- ➡ watsonx_orchestrate_basic_auth_example_integration.md

Develop custom review modes

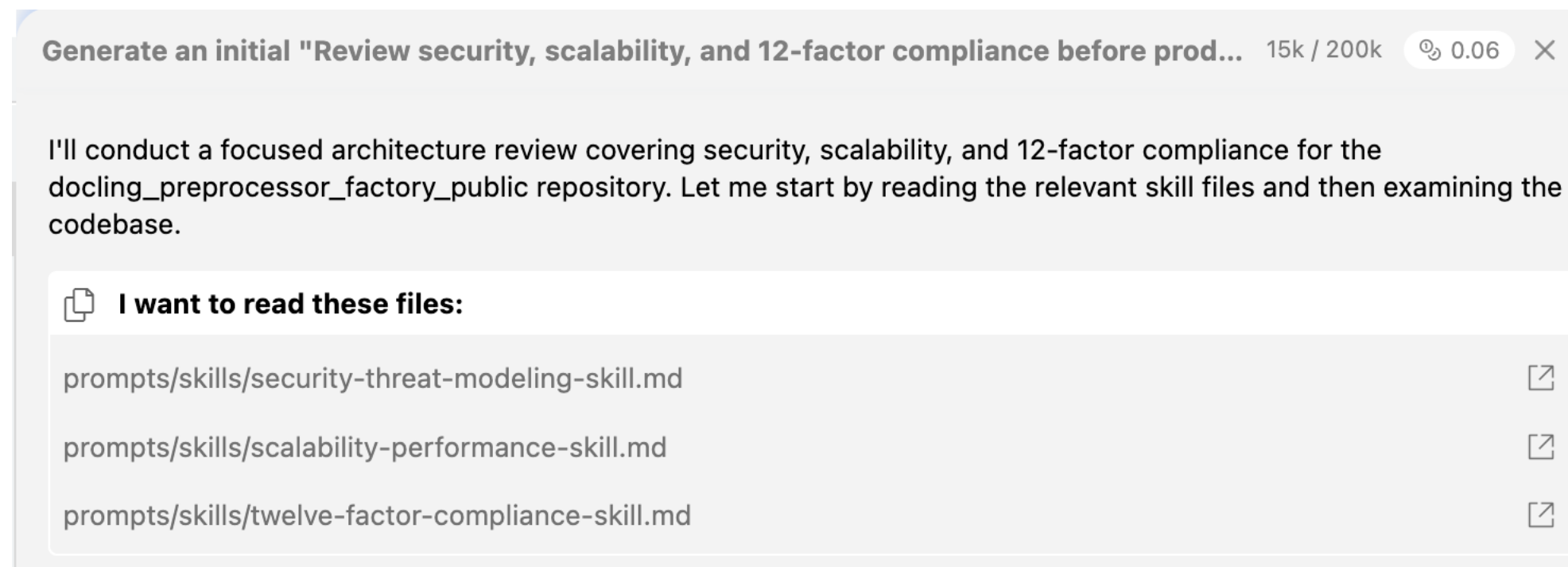


Architecture Review System - using Bob



Example Prompt: Security / Software Architect

Generate an initial review for security, scalability, and 12-factor compliance before production for the GitHub repository source code in the directory @/repos/galaxium-travels-infrastructure-tsuedbro of this project.



General

Account thor

Monthly budget 1

Monthly usage 1
Rem

i Your budget res
on May 1

Language

Language of Bob

Log level

Change the amount tha
by Bob to the output of

Keyboard shortcuts

Configure Keyboard sh

Export Bob settings to

Save all settings and co

Mac rep



Hi, I'm Bob

Ask me questions or let me code for you.

Documentation

☐ Auto-approval disabled

Generate an initial review for security, scalability, and 12-factor compliance before production for the GitHub repository source code in the directory `@/repos/galaxium-travels-infrastructure-tsuedbro` of this project.

12-Factor App Compliance Review

Pre-Production Architecture Review - Executive Summary

Galaxium Travels Infrastructure
Review Date: 2026-04-12
Reviewer: Architecture Review System
Review Scope: Security, Scalability, 12-Factor Compliance

Executive Summary

This comprehensive pre-production review evaluates the Galaxium Travels Infrastructure across three critical dimensions: **Security**, **Scalability**, and **Cloud-Native Readiness (12-Factor)**. The system demonstrates strong foundational architecture with good authentication mechanisms and containerization, but has **critical gaps** that must be addressed before production deployment.

Overall Assessment

Dimension	Score	Status	Readiness
Security	⚠ Moderate Risk	6/10	❌ NOT READY
Scalability	⚠ Moderate	5/10	⚠ LIMITED
12-Factor Compliance	⚠ Fair	72/120 (60%)	⚠ PARTIAL
Overall Production Readiness			❌ NOT READY

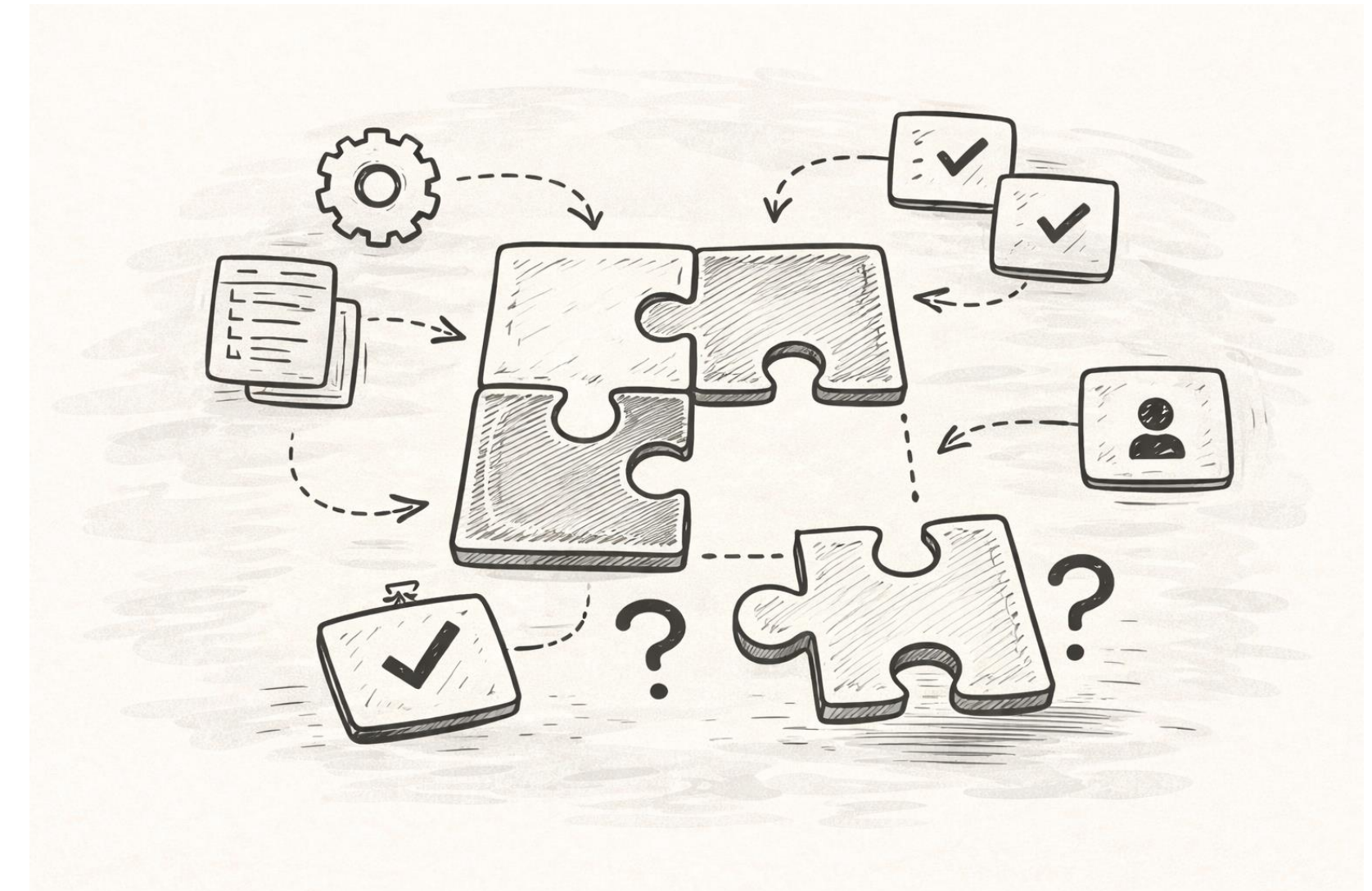
12-Factor Compliance Summary

Factor Compliance Table

Factor	Status	Score	Compliance %	Notes
I. Codebase	✅	10/10	100%	Single repo, multiple deploys
II. Dependencies	⚠	6/10	60%	CRITICAL: Unpinned versions
III. Config	⚠	7/10	70%	CRITICAL: Hardcoded secrets
IV. Backing Services	⚠	7/10	70%	Good abstraction, needs resilience
V. Build, Release, Run	✅	9/10	90%	Clear separation, minor improvements
VI. Processes	⚠	7/10	70%	Stateless backends, session issues
VII. Port Binding	✅	10/10	100%	Self-contained services
VIII. Concurrency	❌	4/10	40%	No scaling mechanisms
IX. Disposability	✅	9/10	90%	Fast startup, graceful shutdown
X. Dev/Prod Parity	⚠	6/10	60%	Database parity critical
XI. Logs	❌	3/10	30%	No structured logging, no aggregation
XII. Admin Processes	❌	4/10	40%	No migrations, no CLI
Overall Score		72/120	60%	FAIR - Significant work required

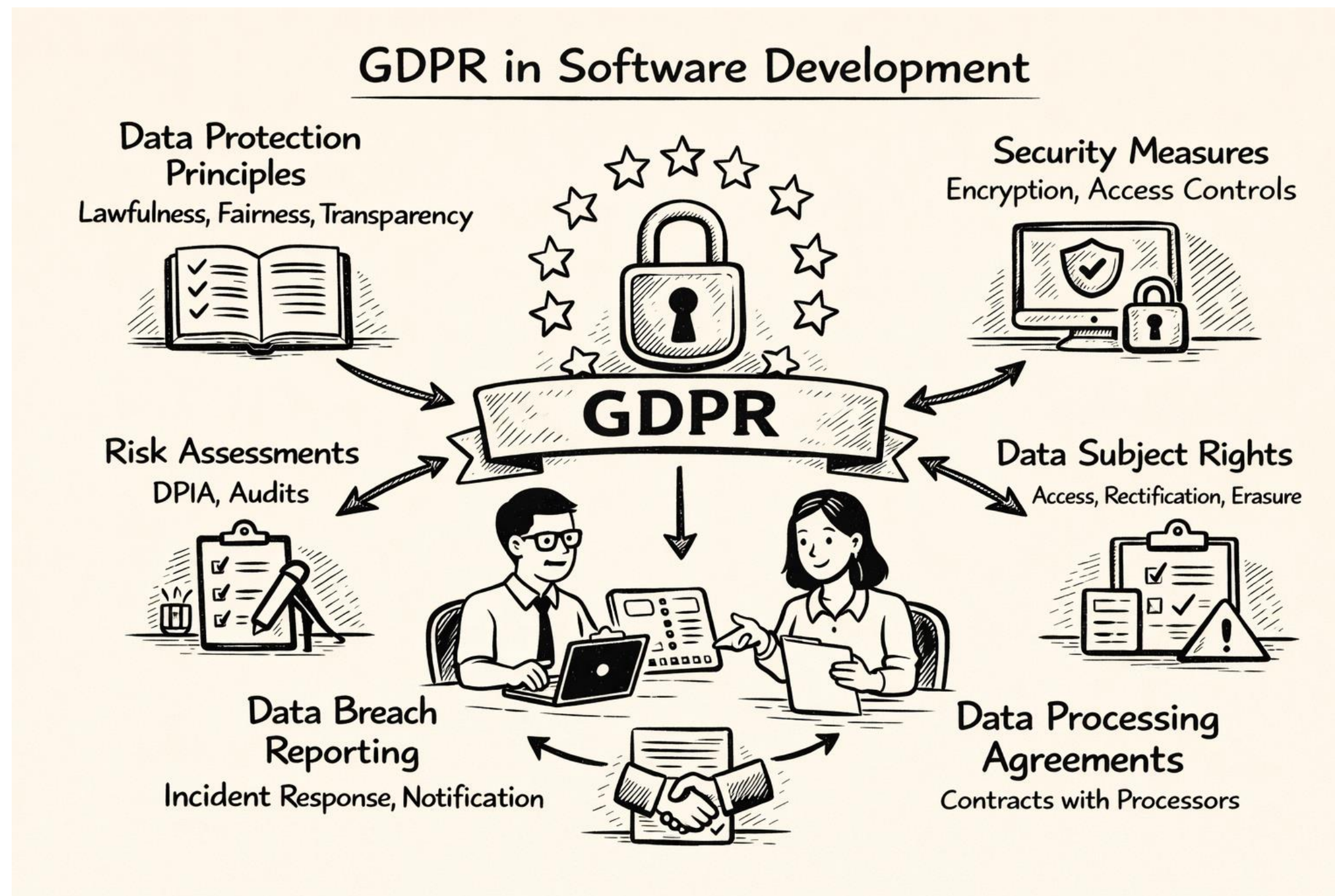
Configuration Gap Detector (🔍)

- Validates configuration fit before execution
- Detects missing Modes or Skills
- Identifies domain, workflow, and governance gaps
- Proposes minimal, targeted extensions
- Keeps configuration modular and up-to-date



Example Prompt: Security and Compliance Officer

Generate a review for DSGVO compliance in Germany before production for the GitHub repository source code in the directory @/repos/galaxium-travels-infrastructure-tsuedb



⌵

🔍

🔗

🔧

📁

👤

📄

🏠

🔄

👤

⚙️

Settings

IBM BOB

Search settings...

General

Chat

Editor

Bob findings

MCP

Modes

Version: 1.0.1

Bob Docs

Report issue

General

Account

Monthly budget

Monthly usage

Your budget res on May 1

Language

Log level

Keyboard shortcuts

Export Bob settings to

zsh - repos

thomassuedbroecker@Mac rep

os %

Recent Tasks

Generate an initial review for security, scalability, and 12-factor compliance before production for the GitHub repository source code in the directory @/repos/galaxium-travels-infrastructure-tsuedbro of this project.

5 minutes ago · 0.76

Auto-approval disabled

Generate a review for DSGVO compliance in Germany before production for the GitHub repository source code in the directory @/repos/galaxium-travels-infrastructure-tsuedbro of this project. Does it work in the USA from a legal perspective?

Architecture ...

review-sdlc-expert-bob-configuration (Workspace)

80%

Hi, I'm Bob

Ask me questions or let me code for you.

Documentation

main

0 0

Live Share

Bob Findings

Go Live

Bob - Settings

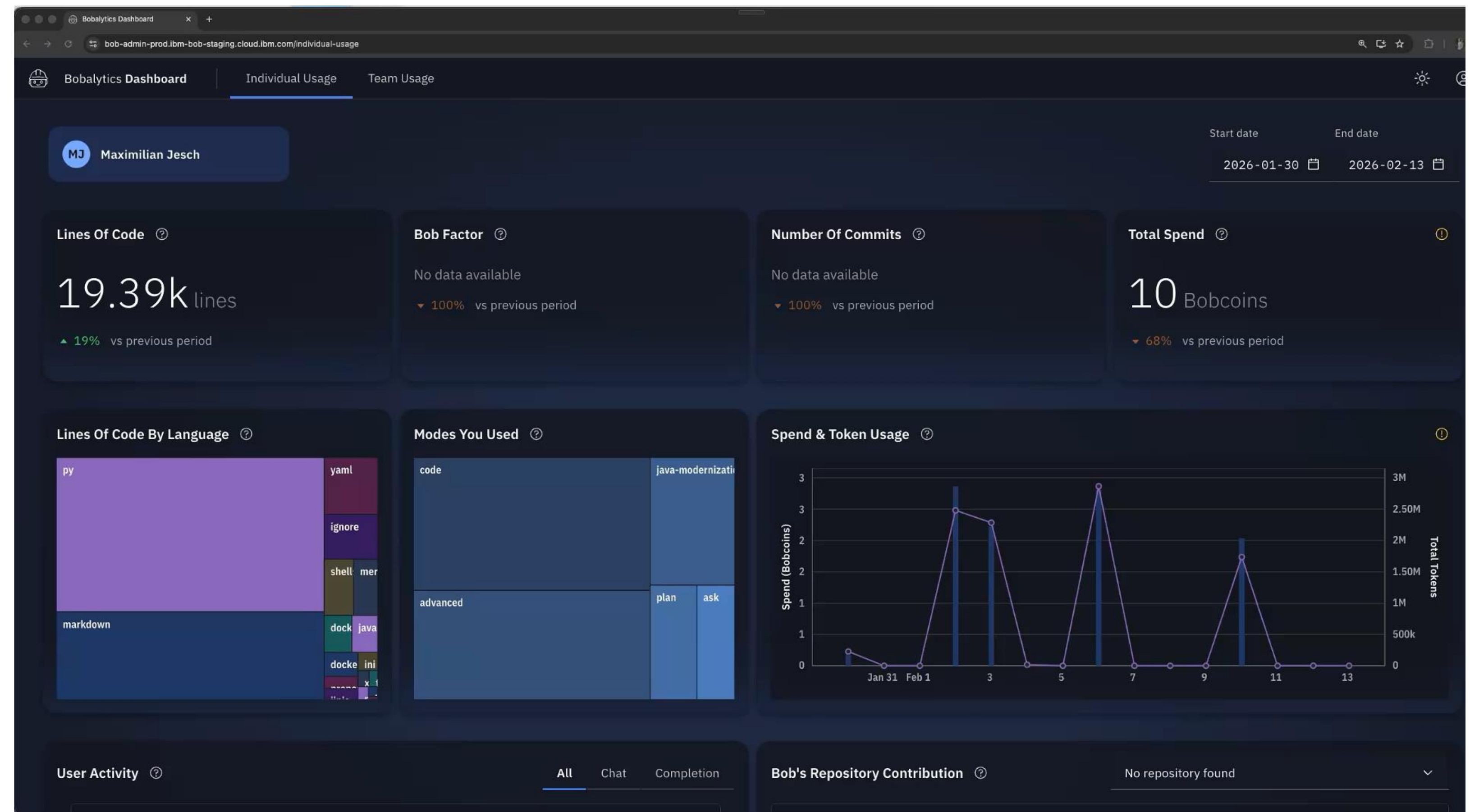
Plan: IBM internal version

Cost optimization

Cost for AI tools is real-time engineering signal that must be actively managed.

But model behavior and resource consumption are often opaque and difficult to control.

IBM Bob provides visibility, control, and optimization of cost across workflows, enabling teams to scale AI usage responsibly.



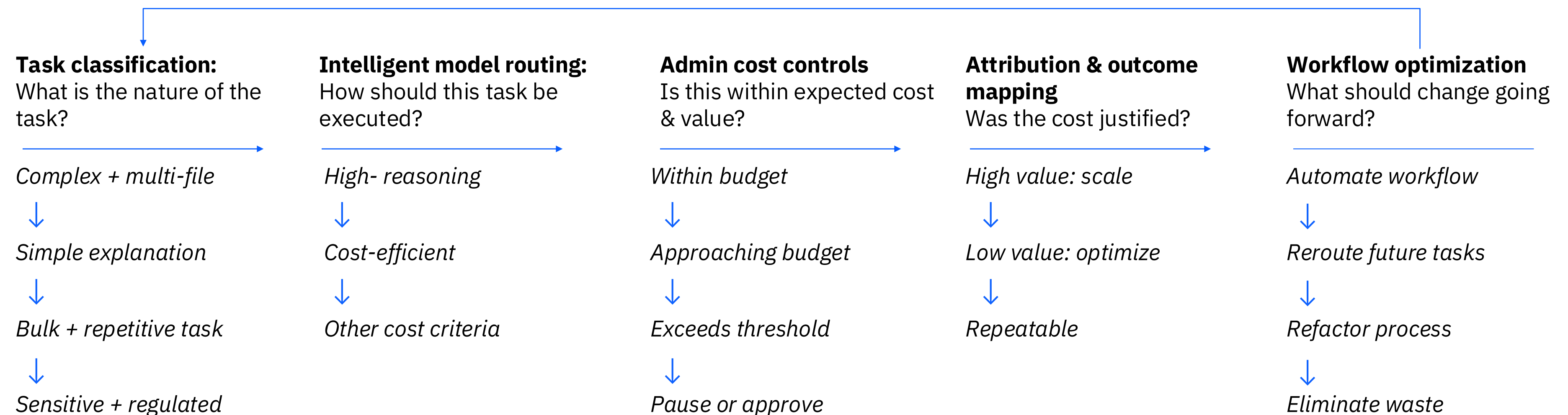
Cost optimization

Cost for AI tools is **real-time** engineering signal that must be actively managed.

But model behavior and resource consumption are often opaque and difficult to control.

IBM Bob provides visibility, control, and optimization of cost across workflows, enabling teams to scale AI usage responsibly.

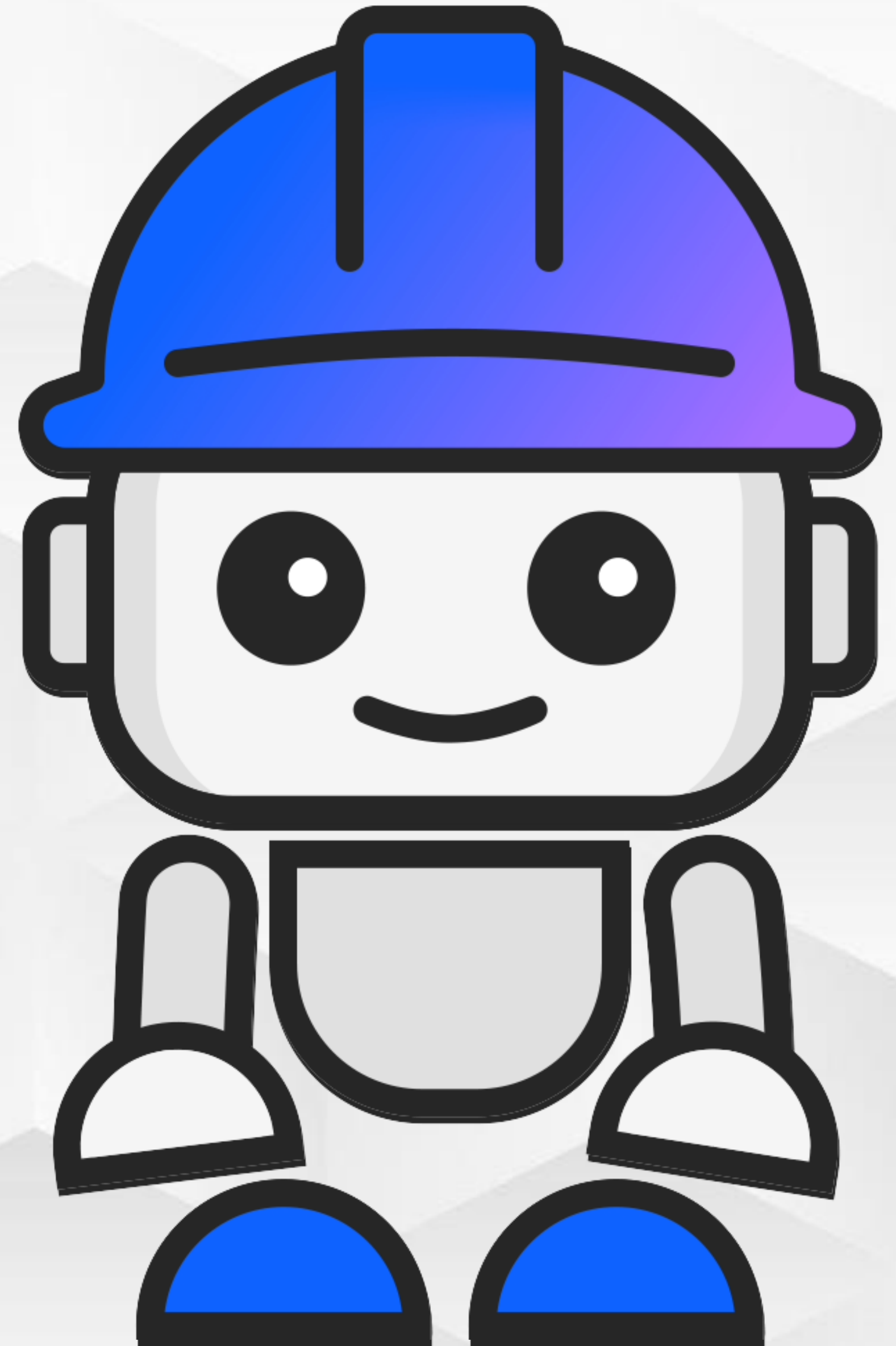
With IBM Bob, you have a decisioning system that optimizes cost at every step



Agent Ops

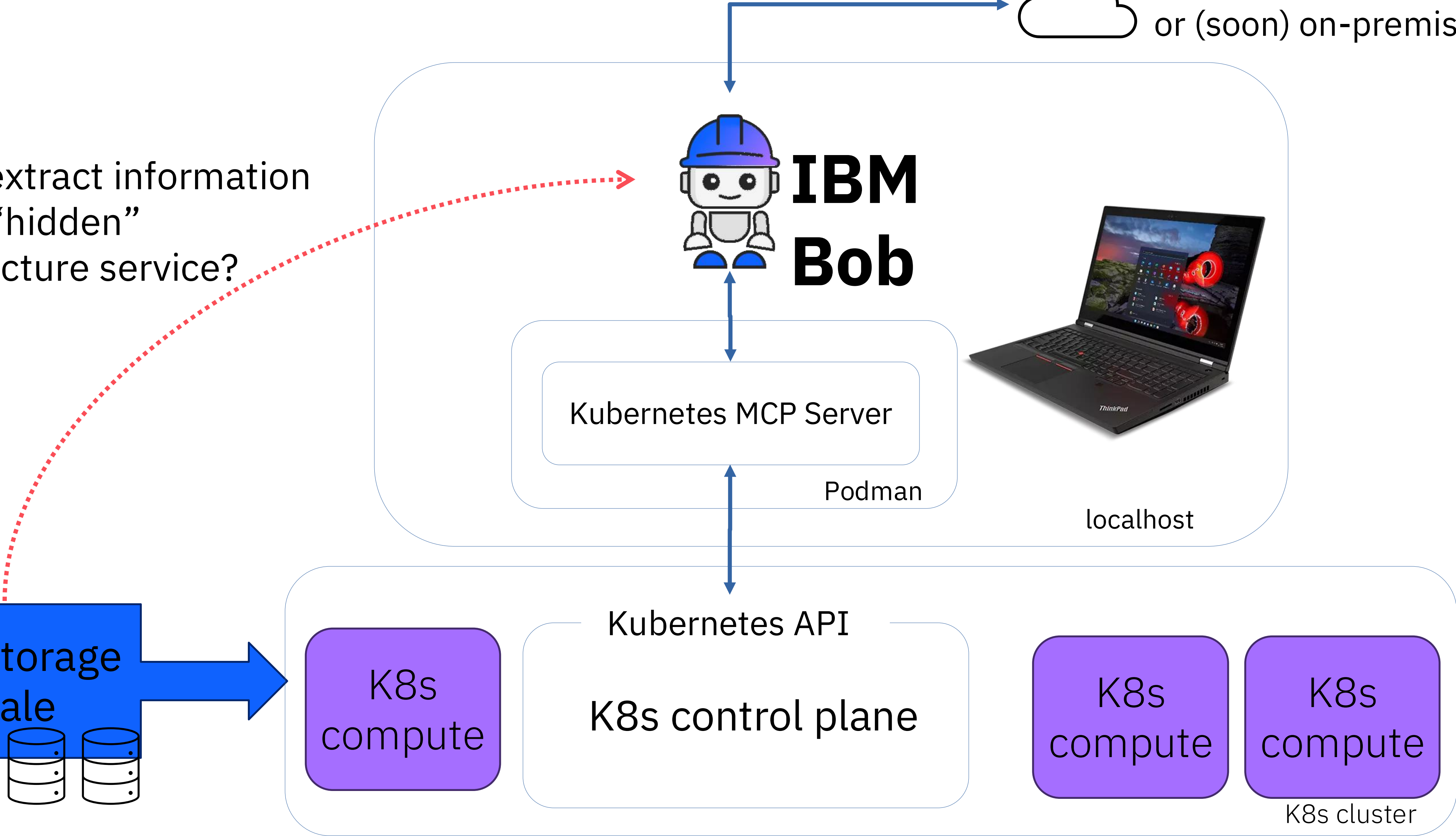
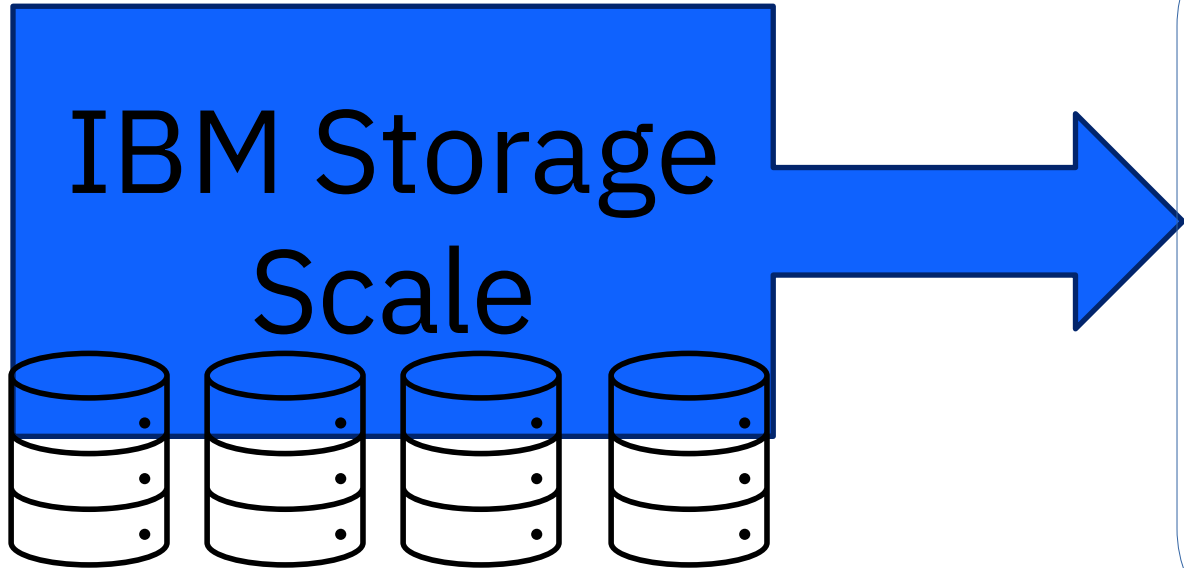
Harald Seipp (IBM)

QAware GmbH Cloud Native Night 09.06.2026



AgentOps with IBM Bob demo setup

Can we extract information about a “hidden” Infrastructure service?



[hseipp@hs-p15g2 fusion]\$

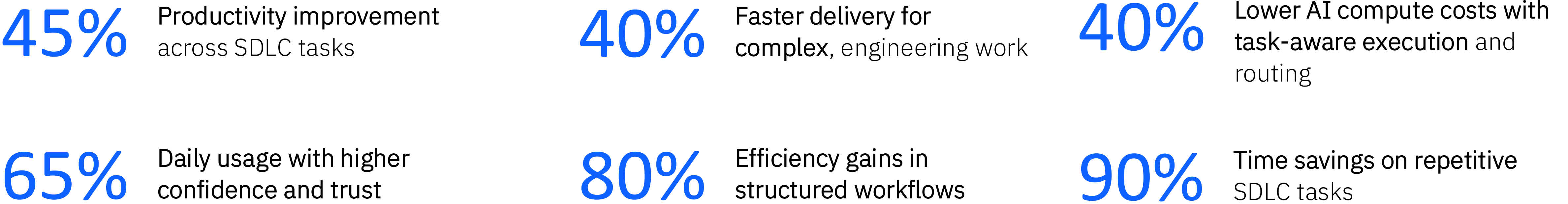
[1] 0: bash*

"hs-p15g2" 17:00 05-Jun-26

What are Bob's strengths?

Delivering AI at scale with ROI requires a unique approach. Bob is designed for that.

Source for ROI metrics: IBM Client 0



An AI partner teams want to use every day

Enable consistent, confident engineering across the SDLC, with an AI partner teams want to adopt

Help teams work faster without lowering quality

Scale software development while maintaining quality

Manage and costs across the full development

Optimize, control, and reduce costs across the SDLC

Catch issues early, reduce rework, and ship with confidence

Detect issues earlier, reduce rework, and deliver with confidence

IBM Bob Premium Packages

Premium packages extend Bob with product-specific capabilities, workflows, and optimizations tailored to each platform.

Each package builds on a common foundation of capabilities designed for enterprise development:

- ✓ Prebuilt workflows & modes to accelerate development processes
- ✓ Domain expertise to optimize tasks even further
- ✓ Connect seamlessly with other IBM tools and systems
- ✓ Domain-specific security features to ensure safe development
- ✓ Other features to improve performance

Premium Packages

IBM Bob Premium Package for Java

Accelerate Java modernization with safer upgrades and migrations.

c

IBM Bob Premium Package for i

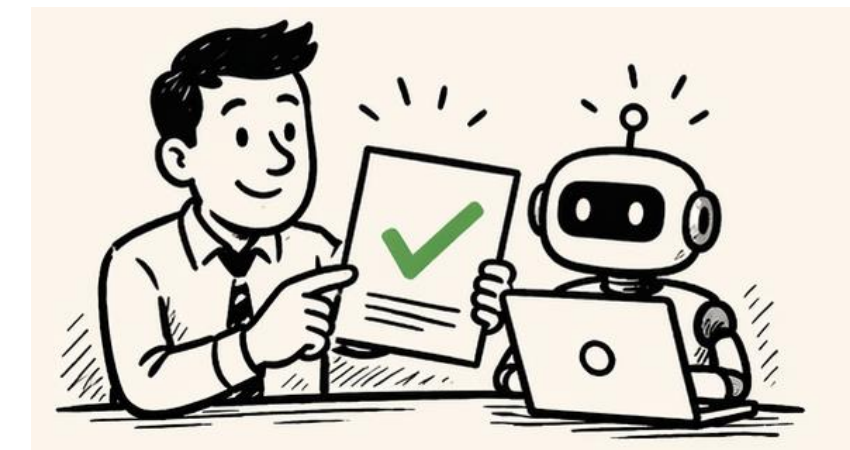
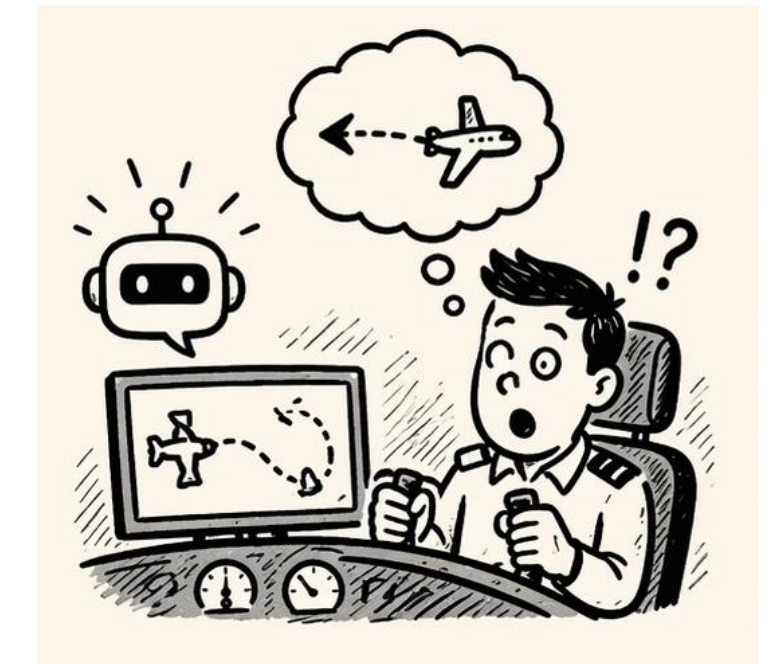
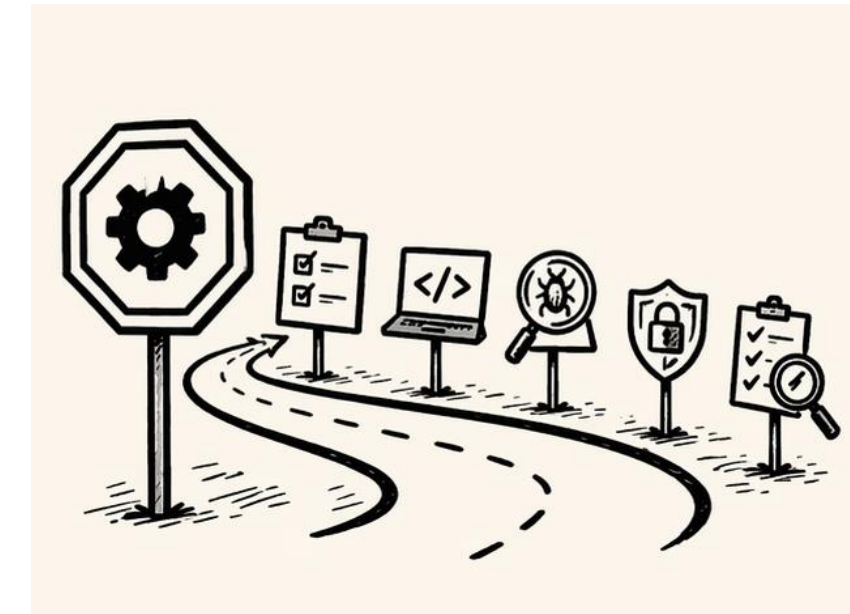
Unlock productivity across IBM i development and modernization.

IBM Bob Premium Package for Z

Optimize mission-critical mainframe applications with platform-aware AI.

Key Learnings & Takeaways

- **SDLC Support Doesn't Stop at Delivery**
Generative capabilities assist across planning, coding, testing, security, and review.
- **Generative AI = Acceleration, Not Autopilot**
AI can draft and analyze, but we must verify correctness, security, and compliance.
- **Human Oversight Remains Essential**
We should understand what we deliver and double-check outputs.
- **Business Value & ROI Depend on Trustworthiness** Real value comes from reliably addressing business intent.
- **Build Trust Through Transparency & Validation**
Documentation, traceability, and validation build confidence.



Thanks

Harald Seipp (IBM)
Thomas Südbröcker (IBM)

